
BANTAY

01 THE CHALLENGE



When help depends on signal, help can arrive too late.

For an older adult living alone, a fall can become dangerous when nobody knows it happened.

⚠️ **THE CORE PROBLEM: detecting is only useful if an alert can follow.**

02 THE SOLUTION

BANTAY keeps watch on the phone itself.

An offline-first safety companion that detects suspicious motion, starts a local countdown, and prepares an emergency SMS.

01

SENSE

Reads motion sensors on-device.

02

CHECK

Evaluates impact, orientation change, and stillness.

03

ALERT

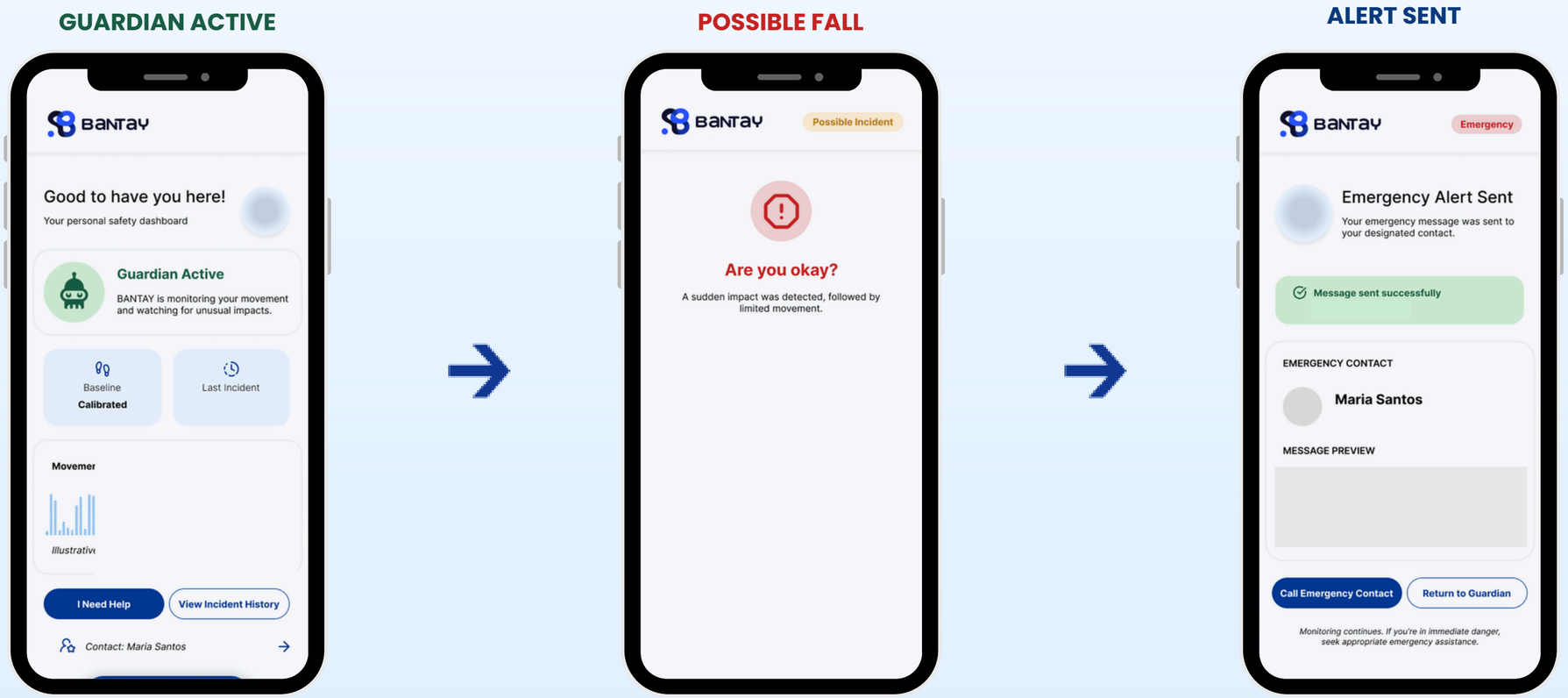
Gives the user a chance to cancel, then sends an SMS.

Designed for a real-world constraint: the cloud can disappear, but the safety flow continues.

03 PRODUCT FLOW

A simple flow in a high-stress moment.

The user stays in control. The safety-critical path does not wait for generative AI.

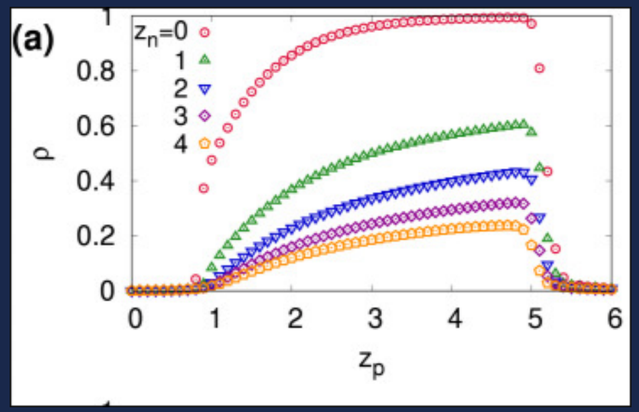


A visible countdown creates a chance to cancel a false alarm before the message is sent.

04 LOCAL AI

Two paths. One non-negotiable rule.

A life-saving path should never wait on a generative model.



Threshold cascade

PATH 1 • SAFETY-CRITICAL

TARGET: <100 ms

Qwen3

PATH 2 • LOCAL INTELLIGENCE

2 s TIMEOUT + FALLBACK

No cloud inference in the critical path. Incident history stays on-device.

05 THE PROOF

Airplane mode is not a failure state.

Our central demo moment: turn connectivity off, then show that the safety flow remains available.

On-device detection

Sensor processing and threshold checks run locally.

Private incident log

History is stored on the phone, available offline.

SMS, not data

SMS can work without mobile data when cellular service is available.

LIVE DEMO



AIRPLANE MODE: ON

No internet. No cloud call. Local detection continues.

06 WHY BANTAY

Resilient by default. Honest about its limits.

Built around a clear promise: remain useful when cloud services are unavailable.

RESILIENT

Core detection and incident flow do not depend on internet access.

PRIVATE

Motion signals and incident history stay on the device by design.

PRACTICAL

Uses a mid-range Android phone and SMS for emergency contact.

WE WILL NOT OVERCLAIM

BANTAY is a safety aid, not a medical device. Fall detection needs real-world validation; phone drops and everyday movement can cause errors.

BANTAY

When the cloud disappears, BANTAY stays on watch.

The safety net that never phones home.

THANK YOU